QALA

Universal Solution Management Operating System

Problem Requirements Document | v2.0 | Universal Edition

Field	Value
Document Type	Problem Requirements Document (PRD) — Universal Edition v2.0
System	Qala Universal Solution Management OS
Classification	Confidential
Status	Draft — For Review
Core Thesis	One OS. Every solution type. Every industry. Every lifecycle stage.
Solution Types	Application, System, Platform, Tool, Toolchain, Development Kit, Library, Package, API, Good, Consumer Packaged Good, Capital Good, Product, Product Line, Service, Managed Service, Professional Service, Financial Instrument, Investment Solution, Capital Solution, Tax Solution, Agricultural Solution, Business Solution, Process Solution, Research Asset, Resource, Asset, Template, Framework, and all generic industry solutions
Capability Pillars	Management + Delivery + Release + Change Management + Quality Management + Development Environment + Configuration Management + Testing Environment + Distribution + Governance Infrastructure
Audience	Product, Engineering, Leadership, Domain Experts, Investors — All Industries

1 Executive Summary

Qala is the world's first Universal Solution Management Operating System. It provides a single, governed, AI-augmented platform that manages the complete lifecycle of any solution — software, hardware, goods, services, financial instruments, agricultural systems, business processes, research assets, or any other form of created value — through one coherent operating model.

The software industry has spent decades perfecting the infrastructure for delivering software: CI/CD pipelines, test management, artefact registries, release management, change control, security scanning. The rest of the economy — the manufacturers, farmers, financiers, consultants, researchers, and public servants who create the overwhelming majority of the world's economic value — has been left behind, managing their solution lifecycles in spreadsheets, disconnected ERPs, ad-hoc project tools, and informal processes.

Qala changes this. By abstracting the universal patterns of solution lifecycle management — create, configure, develop, test, release, distribute, govern, retire — into a single platform with pluggable domain extensions, Qala delivers world-class lifecycle infrastructure to every organisation that creates something of value.

> *Qala's core thesis: every organisation that builds something of value — regardless of type, industry, or scale — deserves the same world-class solution management infrastructure. Qala is that infrastructure: a universal operating system for solution factories of every kind.*

1.1 The Ten Capability Pillars

Qala delivers ten core capabilities that apply universally across every solution type and industry:

Pillar	What It Provides	Universal Example
1. Solution Management	Single source of truth for every solution across its full lifecycle.	One record for a drug formulation, a trading algorithm, a mobile app, a consulting framework, and a crop management platform — all managed identically.
2. Development Environment	Governed, reproducible, version-controlled environments for creating solutions.	Software IDE; formulation lab environment; financial research workbench; agricultural IoT firmware build; service knowledge workspace.
3. Configuration Management	All solution configuration versioned, governed, and drift-detected.	Code configs, formulation parameters, financial model settings, farm sensor calibrations, service SLA definitions.
4. Change Management	Every change governed through risk-scored proposal, approval, and audit.	Code change, ingredient substitution, pricing model recalibration, crop rotation plan update, tax rule modification.
5. Quality Management	Measurement, benchmarking, and enforcement of quality across the lifecycle.	Test coverage, batch quality, fund risk metrics, yield prediction accuracy, client NPS, research reproducibility score.
6. Testing Environment	Isolated, reproducible environments for validating solutions before release.	Software staging; formulation simulation; financial model backtesting; agronomic field trial environment; legal document review.

7. Release Management	Governed promotion through quality gates to distribution.	Software deployment, product launch, fund inception, service go-live, crop variety release, tax product effective date.
8. Distribution Management	Controlled delivery of released solutions to all target channels and consumers.	App store deploy, retail shelf distribution, fund NAV publication, service catalogue publish, regulatory submission.
9. Governance Infrastructure	Policy enforcement, immutable audit, compliance reporting for all regimes.	SOC 2 for software, GMP for pharma, MiFID II for financial products, GDPR across all, SLSA for builds.
10. AI & Intelligence	Embedded AI Agent providing recommendations, predictions, and automation.	Test generation for software; quality prediction for CPG; yield optimisation for agriculture; risk scoring for financial products.

1.2 The Universal Solution Type Taxonomy

Qala manages every type of solution through one unified model. The taxonomy is organised into ten solution categories:

Category	Solution Types	Managed By
Software & Technology	Application, System, Platform, API, Microservice, Firmware	Development teams, DevOps, Platform Engineering
Tooling & Ecosystem	Tool, Toolchain, Development Kit (SDK), Library, Package, Framework, Reference Architecture	Platform teams, Open Source maintainers, SDK authors
Physical Goods	Good, Consumer Packaged Good (CPG), Capital Good, Product, Product Line, Component, Assembly, Hardware	CPG companies, Manufacturers, OEMs
Services	Service, Managed Service, Professional Service, Consulting Service, Subscription Service, Support Service	Service businesses, Consultancies, Managed Service Providers
Financial Solutions	Financial Instrument, Investment Solution, Capital Solution, Insurance Product, Tax Solution	Banks, Asset Managers, Insurance Firms, Tax Product Companies
Agricultural Solutions	Agricultural Solution, Crop Management System, Livestock System, Supply Chain Platform, AgTech Platform	Agribusinesses, Farming Cooperatives, AgTech Companies
Business & Operational	Business Solution, Process Solution, Operational Solution, Workflow Solution, Transformation Programme	Enterprises, Operations teams, Digital Transformation leads
Research & Academic	Research Asset, Dataset, Experimental Framework, Computational Model, Publication Pipeline	Research Institutions, Academia, R&D Labs
Resources & Assets	Resource, Asset, Template, Standard, Pattern, Reference Architecture	Any organisation — reusable foundational artefacts
Platform & Ecosystem	Platform, Marketplace, Ecosystem, Multi-Sided Network, Developer Ecosystem	Platform companies, Marketplace operators, Ecosystem orchestrators

2 The Universal Problem Landscape

The problems Qala solves are universal. Every organisation managing solutions — regardless of type — faces the same structural dysfunctions: fragmented tooling, absent lifecycle governance, ungoverned change, poor quality visibility, reactive security, and no structured pathway from creation to distributed value.

2.1 The Root Cause: Every Sector Reinvents the Same Wheel

Walk into a pharmaceutical R&D team and you find formulations managed in Excel, change requests in email, regulatory submissions assembled manually from 12 different systems, and quality records stored in a LIMS that cannot talk to the PLM that cannot talk to the ERP. Walk into a fintech startup and you find code in GitHub, builds in Jenkins, security scanning in Snyk, releases managed in Jira, and compliance evidence assembled in spreadsheets every six months.

Walk into a farm cooperative and you find sensor data in an IoT platform, yield records in an agri-ERP, logistics in a TMS, traceability in a paper ledger, and nobody able to answer a regulator's question about where this batch of produce came from within less than three days.

Walk into a consulting firm and you find methodologies in PowerPoint decks, engagement knowledge in SharePoint folders nobody can find, client outcomes in a CRM that was never designed for outcome tracking, and senior partners as the sole carriers of institutional knowledge.

Different industries. Different tools. The same problem: no unified, governed, intelligent lifecycle management platform for the solutions they create and deliver.

Sector	Current Toolchain	Coordination Cost	Missing Capability
Software	GitHub + Jenkins + Snyk + Artifactory + Jira + Datadog + 20 more	30–50% of engineering time on toolchain vs. product	Unified governance, hermetic builds, continuous compliance
CPG / Manufacturing	SAP PLM + LIMS + QMS + ERP + Regulatory + Spreadsheets	40–60% of ops management time on coordination	Versioned formulations, recall traceability, change control
Financial Services	Bloomberg + Risk engines + Compliance tools + ERP + audit prep	35–50% of product ops time on compliance and governance	Versioned product specs, automated compliance, audit trail
Agriculture	Farm ERP + IoT platform + Logistics TMS + Traceability system	30–45% of operations on data reconciliation	Unified traceability, precision data intelligence, provenance
Professional Services	CRM + SharePoint + Billing + Project tools + Knowledge base	25–40% of service delivery on coordination overhead	Knowledge versioning, outcome benchmarking, methodology mgmt
Research / Academia	LIMS + Data repos + Analysis tools + Publication tools + Code repos	40–60% on reproducibility and provenance management	Reproducible pipelines, dataset versioning, experiment attestation
Legal / Tax	Document mgmt + Billing + Compliance + Tax engine + ERP	30–50% on manual regulatory update processing	Rule versioning, regulatory change automation, audit trail

Government / Public	Multiple legacy systems + Manual processes + Audit spreadsheets	50–70% on governance and compliance overhead	Unified solution governance, automated compliance, public audit

2.2 Problem Cards — Software and Technology Solutions

P-SW-001 — Non-Reproducible Builds and Environment Inconsistency

Persona: Developer, DevOps Engineer, Build Engineer — all software organisations

Context: Every developer runs a slightly different local environment. CI runs in a different environment again. Builds that pass locally fail remotely. Dependencies drift. "It works on my machine" is endemic.

Pain Point: Enormous time wasted debugging environment differences rather than building solutions. Security scanning of what was actually deployed is impossible without reproducible builds. Supply chain attacks via dependency tampering are undetectable.

Business Impact: Industry surveys report 30–50% of engineering time consumed by environment and toolchain issues. SLSA research shows that over 80% of software supply chain attacks are undetectable without cryptographic build attestation.

P-SW-002 — Security Is Bolted On, Not Built In

Persona: Developer, Security Engineer, CISO — all software organisations

Context: Security review happens at the end of the development cycle — in a penetration test before launch, or after a breach. Developers have no security feedback during development. Known CVEs in dependencies can persist for months.

Pain Point: Security findings arrive too late to fix cheaply. Developers resist last-minute changes. Vulnerability debt accumulates. Supply chain attacks compound an already compromised posture.

Business Impact: IBM Cost of a Data Breach: vulnerabilities found in production cost 6–15× more than those found in development. SolarWinds, Log4Shell, and XZ Utils demonstrate the scale of undetected supply chain attacks.

P-SW-003 — No Governed Release and Deployment Process

Persona: Release Manager, Developer, Operations Engineer

Context: Production deployments are manual, inconsistent, and ungoverned. No enforceable gate checklist. Changes go to production without test sign-off, security clearance, or performance validation. Rollbacks are slow and painful.

Pain Point: Deployment is the riskiest event in software delivery but the least governed. Post-deployment incidents are common. Mean time to recovery is days, not minutes. Change failure rate is 15–45% across the industry.

Business Impact: DORA Research: elite teams deploy 208× more frequently with 2,604× faster recovery. The difference is entirely in governance and automation — not team skill or size.

P-SW-004 — Tool, Library and Package Lifecycle Is Ungoverned

Persona: Developer, Platform Engineer, Open Source Maintainer

Context: Libraries, packages, SDKs, and internal tools are created, updated, and deprecated without consistent lifecycle governance. Version proliferation creates dependency hell. Security vulnerabilities in shared libraries persist for years because no one tracks consumer impact.

Pain Point: No registry of authorised tools. No governed deprecation. No mandatory security scanning before publication. No reliable signal of library maintenance status or security posture.

Business Impact: Log4Shell affected hundreds of millions of systems because a ubiquitous library had an undetected zero-day and no tooling existed to identify all consumers and coordinate remediation. This class of problem is structural, not accidental.

2.3 Problem Cards — Physical Goods, CPG and Manufacturing

P-GOOD-001 — Formulation and Specification Version Chaos

Persona: Product Developer, Regulatory Affairs Manager, Quality Manager — CPG, Pharma, Food, Manufacturing

Context: Product formulations, Bills of Materials, ingredient specifications, and manufacturing parameters live in spreadsheets, email threads, and disconnected PLM/ERP systems. Multiple versions circulate simultaneously. Changes are made without formal governance.

Pain Point: Regulatory submissions reference outdated formulations. Production runs use wrong specifications. Audits find undocumented changes. Recall investigations cannot identify the causative version.

Business Impact: FDA warning letters frequently cite inadequate change control as the primary finding. Product recalls from formulation management failures cost $10M–$600M per incident in direct costs alone, before regulatory penalties and brand damage.

P-GOOD-002 — Recall Readiness Is Absent or Reactive

Persona: Quality Manager, Operations Director, Regulatory Affairs — CPG, Food, Pharma, Medical Devices

Context: When a product defect or contamination is discovered, tracing affected batches through the supply chain — from raw material supplier through manufacturing to retail shelf — requires days or weeks of manual investigation across disconnected systems.

Pain Point: Batch-level traceability data exists across 5–10 separate systems that cannot be queried coherently. The time between defect discovery and recall completion is extended, increasing consumer harm, regulatory exposure, and litigation risk.

Business Impact: The FDA estimates each day of delay in a Class I recall costs $1M+ in compounding regulatory and litigation risk. The 2011 European E. coli outbreak caused 50 deaths and $1.3B in economic damage partly from traceability failures.

P-GOOD-003 — Product Line Governance Across Variants Is Impossible

Persona: Product Line Manager, R&D Director — CPG, Manufacturing, Pharma, Electronics

Context: A product line of 50 SKUs shares common formulation components, specifications, and regulatory approvals. When a core component changes, identifying which SKUs are affected, updating their specifications, and re-validating the affected variants requires manual cross-referencing across all 50 records.

Pain Point: No platform models a Product Line as a managed entity with governed relationships to its member products. Changes cascade unpredictably. Impact assessment for a component change takes weeks.

Business Impact: Unmanaged product line complexity is responsible for 23% of unnecessary product recalls (McKinsey CPG operations research) because affected variants are not identified in time.

2.4 Problem Cards — Financial, Investment and Capital Solutions

P-FIN-001 — Financial Product Lifecycle Is Ungoverned

Persona: Product Manager, Risk Manager, Compliance Officer — Banks, Asset Managers, Insurance firms

Context: Financial instruments, investment funds, structured products, and insurance policies are created and modified without formal lifecycle governance. Product term changes, risk model updates, and pricing parameter modifications happen through informal processes.

Pain Point: No version control for financial product specifications. Parameter changes undocumented. Model updates untracked. Audit trails incomplete. Regulatory examiners find governance gaps on every examination.

Business Impact: MiFID II, Basel III, IFRS 9, and Solvency II all require full governance audit trails for financial product changes. Regulatory fines for inadequate product governance range from $1M to $1B+ for systemically important institutions.

P-FIN-002 — Capital and Investment Solution Tracking Across Vehicles

Persona: Fund Manager, Portfolio Manager, Capital Allocator — Asset Management, Private Equity, Family Offices

Context: An investment manager operating 15 funds, 30 separately managed accounts, and a co-investment programme has no unified platform governing the lifecycle of each investment solution — from mandate design through portfolio construction to investor reporting.

Pain Point: Each investment vehicle is governed by different tools, different processes, and different personnel. Mandate drift goes undetected. Version-controlled investment policies do not exist. Investor reporting draws from inconsistent sources.

Business Impact: The 2010 Flash Crash and multiple subsequent risk incidents were partially attributed to undetected model drift in algorithmic trading systems — a direct consequence of absent version control and validation governance for financial solutions.

P-FIN-003 — Tax Solution Changes Are Manual and Risky

Persona: Tax Product Manager, Tax Technologist, Compliance Team — Tax Software Companies, In-house Tax Departments

Context: Tax products must be updated in response to legislative changes across dozens of jurisdictions simultaneously. These updates are manual, ungoverned, and race against hard legislative effective dates. Multiple conflicting product versions reach production.

Pain Point: No version control for tax rule sets. Jurisdiction variants tracked manually. Regulatory effective dates missed. Clients on different versions receive different compliance outcomes for identical inputs.

Business Impact: Tax software errors affect millions of taxpayers annually. The IRS receives approximately 1.2M amended returns each year, a significant proportion attributable to tax software calculation errors from ungoverned product changes.

2.5 Problem Cards — Agricultural and Agribusiness Solutions

P-AGRI-001 — Farm-to-Consumer Traceability Is Broken

Persona: Food Producer, Supply Chain Manager, Retailer, Regulator — Agriculture, Food Production, Logistics

Context: When a food safety incident occurs, tracing from the affected consumer product back through distribution, processing, and farming to the source field or animal requires 2–7 days of manual investigation across 4–8 disconnected systems.

Pain Point: Farm management software, processing plant MES, logistics TMS, and retail WMS do not exchange event data. Batch-level provenance rarely survives more than one handoff in the chain.

Business Impact: The economic cost of a major food recall averages $10M direct and $30M brand damage. The 2018 romaine lettuce E. coli outbreak cost $1.5B in economic losses and affected 62 people because traceability could not quickly narrow the contamination source.

P-AGRI-002 — Agricultural Solutions Are Not Lifecycle-Managed

Persona: Agricultural Technology Provider, Farm Operations Manager, AgTech Investor

Context: Agricultural solutions — precision irrigation systems, crop disease detection platforms, soil health monitoring tools, livestock management systems — are built and deployed without structured solution lifecycle management. Farmers run outdated firmware on sensors, deprecated software on tablets, and unvalidated models in decision support tools.

Pain Point: No governed versioning for agri-solutions. No update pipeline for field-deployed hardware. No quality benchmarking for agronomic AI models. No change governance for platform updates that affect yield-critical decisions.

Business Impact: The potential productivity improvement from well-governed precision agriculture deployment is 15–25% yield increase (McKinsey). The gap between potential and achieved is primarily a governance and lifecycle management failure, not a technology limitation.

2.6 Problem Cards — Services, Consulting and Professional Solutions

P-SVC-001 — Service Intellectual Capital Has No Lifecycle Management

Persona: Partner, Principal, Knowledge Manager — Consulting, Coaching, Professional Services

Context: The methodologies, frameworks, engagement playbooks, and accumulated client insights that constitute a professional services firm's intellectual capital exist in PowerPoint decks, SharePoint folders, and individual practitioners' heads. There is no versioned, governed, searchable repository.

Pain Point: When a principal leaves, their methodology goes with them. Engagements reinvent approaches that exist elsewhere in the firm. Quality varies wildly by team. Client knowledge cannot be leveraged across relationships.

Business Impact: Consulting firms estimate 30–50% of their intellectual capital is "dark knowledge" — unrecorded and inaccessible. Annual cost of replacing a senior consultant who leaves: 50–200% of annual compensation in lost institutional knowledge.

P-SVC-002 — Managed Service Specifications Are Not Governed Products

Persona: Managed Service Provider, Service Architect, Customer Success Manager

Context: Managed service offerings — their scope, SLAs, runbook definitions, support tiers, and contractual terms — are designed once and then drift informally. Changes to service specifications are undocumented. Customers on different "versions" of nominally the same service receive different outcomes.

Pain Point: Service specification drift is invisible. SLA changes propagate inconsistently to customers. Quality benchmarking against service specifications is impossible without formal versioning.

Business Impact: Gartner estimates 40% of managed service contract disputes arise from scope ambiguity caused by undocumented service specification changes after contract signature.

2.7 Problem Cards — Business, Process and Operational Solutions

P-BIZ-001 — Business Solutions Have No Lifecycle After Launch

Persona: Digital Transformation Lead, Operations Director, Business Analyst

Context: A business transformation programme designs a new operating model, deploys it, and then the platform team disbands. Six months later, the "as-operated" process has diverged substantially from the "as-designed" specification. Nobody knows what the current state of the solution actually is.

Pain Point: No version control for process models. No change governance after launch. No quality benchmarking against the designed outcomes. No mechanism to detect drift between the designed and operated solution.

Business Impact: McKinsey estimates 70% of digital transformation programmes fail to achieve their intended business outcomes. The primary cause is absence of structured lifecycle management — the solution is designed and launched but never governed.

2.8 Problem Cards — Research, Academic and Scientific Solutions

P-RES-001 — Research Is Not Reproducible at Scale

Persona: Research Scientist, Principal Investigator, Research Institution, Regulatory Submitter

Context: Computational research — data analysis pipelines, simulation models, statistical processing — cannot be reproduced because the exact software environment, dataset version, and pipeline that produced the original result were never formally captured.

Pain Point: Published results cannot be independently verified. Regulatory submissions (FDA, EMA) for computational evidence are questioned. Collaborators cannot build on each other's work. The research record is unreliable.

Business Impact: A 2021 Nature survey found 70% of researchers have failed to reproduce another scientist's published results. The annual cost of irreproducible research in the US is estimated at $28B (Freedman et al., PLOS Biology).

2.9 Problem Cards — Resources, Assets, Templates and Frameworks

P-RES-002 — Organisational Resources Have No Governed Lifecycle

Persona: Architect, Standards Body, Knowledge Manager, Platform Team

Context: Reference architectures, design system assets, policy templates, security frameworks, and compliance checklists are created once and published without subsequent lifecycle governance. Consumers cannot know if the resource they are using is current, deprecated, or superseded.

Pain Point: Resources drift out of alignment with current standards. Deprecated templates remain in active use. No notification mechanism for consumers when a resource they depend on is updated.

Business Impact: Organisations using ungoverned architectural resources accumulate technical debt in proportion to the staleness of those resources. The cost of remediation compounds with every new system built on a deprecated reference.

3 Addressable Market and Persona Analysis

Qala's addressable market is not developer tooling — it is the entire solution economy. Every organisation that creates, manages, or delivers a solution of any type is a potential Qala customer.

3.1 Total Addressable Market

Segment	Solution Types	Global Organisations	Market Context
Software & Technology	Applications, APIs, Systems, Platforms, Microservices	8M+ organisations	$850B DevOps/DevTools TAM. Growing 18% CAGR.
Consumer Packaged Goods	Goods, Products, Product Lines, CPG formulations	500K+ organisations	$4.3T CPG industry. PLM and QMS software $12B TAM.
Manufacturing	Capital Goods, Products, Components, Assemblies	500K+ organisations	$2.3T manufacturing. Digital manufacturing software $20B TAM.
Financial Services	Instruments, Funds, Loans, Insurance, Tax Products	150K+ organisations	$400B+ financial technology TAM. Regtech $21B and growing.
Agriculture & Agribusiness	Agricultural Platforms, Crop Systems, Supply Chain	1M+ organisations	$22B AgTech market. Growing 12% CAGR.
Professional Services	Consulting Products, Service Frameworks, Methodologies	2M+ organisations	$600B professional services industry. 5–8% on tooling.
Healthcare & Life Sciences	Medical Solutions, Drug Formulations, Clinical Tools	200K+ organisations	$30B+ HealthTech and $15B+ Life Sciences software TAM.
Research & Academia	Research Assets, Datasets, Computational Models	500K+ institutions	$15B+ research software market.
Government & Public Sector	Public Services, Policy Solutions, Regulatory Products	500K+ agencies	$500B+ government IT spend annually.
Legal & Tax	Tax Products, Legal Templates, Compliance Frameworks	200K+ organisations	$35B legal tech market. $15B tax technology market.
Independent Professionals	Tools, Frameworks, Personal Products, Freelance Deliverables	100M+ individuals	$1.5T global freelance economy.

3.2 Universal Persona Map

Persona	Solutions They Manage	Primary Qala Value Proposition

Software Developer	Applications, APIs, Microservices, Libraries, Packages, SDKs	Hermetic builds, integrated security, SDE provisioning, governed CI/CD, unified quality dashboard.
Platform / DevOps Engineer	Platforms, Toolchains, Infrastructure, Development Environments	Solution Factory governance, tool registry, environment-as-code, drift detection, vendor integration management.
CPG Product Developer	Consumer goods formulations, product lines, variant management	Formulation versioning, ingredient CCR workflow, regulatory gate, batch traceability, recall readiness.
Manufacturing Engineer	Capital goods, assemblies, production specifications, BOMs	BOM versioning, specification change control, quality benchmarking, supplier traceability.
Financial Product Manager	Investment products, financial instruments, pricing models	Versioned product specs, governance for parameter changes, automated MiFID II / SEC compliance, audit trail.
Portfolio / Fund Manager	Investment vehicles, mandates, portfolio construction methodologies	Fund lifecycle management, mandate change governance, model version control, investor report provenance.
Agricultural Operations Manager	Crop management systems, IoT platforms, supply chain solutions	Solution lifecycle for agri-platforms, IoT SDE, seasonal release cadence, farm-to-consumer traceability.
Professional Services Lead	Consulting methodologies, engagement deliverables, service frameworks	Methodology versioning, AI-powered knowledge retrieval, outcome benchmarking, engagement quality governance.
Research Scientist / PI	Datasets, computational models, experimental frameworks, publications	Reproducible research SDEs, dataset versioning and attestation, experiment result provenance.
Compliance / Regulatory Officer	Compliance frameworks, audit evidence, regulatory products	Automated compliance evidence for any framework, immutable audit trail, continuous compliance posture.
Tax Product Manager	Tax rule sets, jurisdiction variants, regulatory update products	Rule set versioning, effective-date release scheduling, multi-jurisdiction variant management, change audit trail.
Operations Director	Business solutions, process frameworks, workflow automations	Business solution lifecycle management, process version control, outcome measurement, transformation governance.
Open Source Maintainer	Libraries, packages, frameworks, development tools	Package lifecycle governance, dependency security, SBOM generation, consumer notification on breaking changes.
Independent Builder / Freelancer	Personal tools, freelance deliverables, consumer products	Personal solution factory, SDE templates, portfolio management, signed artefacts, distribution management.

4 Opportunity Statements

For each identified problem domain, Qala creates a clear, measurable opportunity to deliver structured solution lifecycle management at a scale and quality that has never previously been accessible outside the largest technology organisations.

OPP-001 — Single OS for Every Solution Type

Qala provides one platform — one data model, one governance engine, one audit trail, one AI agent — for every solution type an organisation manages. A CPG company manages its formulations, its manufacturing tools, its distribution software, and its retailer service agreements all in one governed platform. A financial firm manages its investment products, its pricing models, its compliance frameworks, and its internal tooling in one system. One platform. Zero integration overhead between solution types.

Success Metrics: Elimination of cross-tool integration overhead. 360° visibility across the entire solution portfolio. Platform becomes the operating system of the organisation.

OPP-002 — Universal Development Environment Management

Qala's Solution Development Environment (SDE) is the universal concept for the workspace in which any solution is created. A software SDE provisions a hermetic build container. A formulation SDE provisions a simulation workspace with versioned chemistry tools. A financial research SDE provisions a quantitative modelling workbench with data access and backtesting tools. A service SDE provisions a knowledge management workspace. All governed, versioned, reproducible, and governed through the same SDE lifecycle model.

Success Metrics: Onboarding time from weeks to hours for every solution type. Environment-caused failures eliminated. Every SDE reproducible from version-controlled definition.

OPP-003 — Governed Change Management for Every Change Type

Every change to every solution — a code commit, an ingredient substitution, a fund mandate update, a tax rule modification, a process redesign, a crop management algorithm update, a service SLA change — flows through Qala's Change Control Request workflow. Risk-scored, routed to the appropriate approver or CCB, implemented, validated, and immutably recorded. No change escapes governance. No audit surprise is possible.

Success Metrics: Zero ungoverned changes across all solution types. Regulatory audit findings related to change control reduced by 90%+. Emergency change resolution from days to hours.

OPP-004 — Quality Management Across Every Solution Domain

Quality management in Qala is not "test coverage for software." It is a universal quality intelligence framework: software test pass rates, pharmaceutical batch quality metrics, financial model risk-adjusted performance, agricultural yield prediction accuracy, service delivery NPS, research reproducibility scores — all managed through the same quality benchmarking engine with AI-driven insights, trend analysis, and cross-domain learning.

Success Metrics: 50%+ reduction in quality-related incidents. AI predicts quality failures before they occur. Cross-domain quality learning surfaces best practices across solution types.

OPP-005 — Hermetic Testing Environments for Every Solution Type

Qala provisions testing environments for any solution type: software staging environments, formulation simulation environments, financial model backtesting sandboxes, agronomic field trial environments, service delivery pilot environments. All testing environments are version-controlled, ephemeral, isolated from production, and traceable to the solution version being tested.

Success Metrics: 100% of solutions tested in governed, reproducible environments before release. Testing environment drift eliminated. Test results fully traceable to environment version and solution version.

OPP-006 — Universal Release and Distribution Management

Qala's release management model applies to every solution type: software deployment gates, pharmaceutical regulatory approval gates, fund inception gates, agricultural solution seasonal release gates, service catalogue publish gates, tax product effective-date gates. Every release has a gate checklist, an attestation record, and a distribution record. Rollback is always available.

Success Metrics: Change failure rate reduced by 70%+ across all solution types. 100% of releases have verifiable gate evidence. Rollback or recall achievable within defined SLAs for every solution type.

OPP-007 — Continuous Compliance Across Every Regulatory Regime

Qala's immutable audit trail and modular compliance engine generate evidence for every regulatory framework — SOC 2, ISO 27001, GDPR, CCPA, FDA GMP, 21 CFR Part 11, REACH, MiFID II, Basel III, IFRS 9, Solvency II, SLSA, FedRAMP — continuously, not periodically. Compliance is a permanent operational state, not a pre-audit scramble.

Success Metrics: Compliance preparation time from weeks to hours for all supported frameworks. Audit findings reduced by 80%+. Compliance as a competitive advantage for any solution type in any industry.

OPP-008 — AI Intelligence Across the Universal Solution Lifecycle

Qala's AI Agent is domain-aware. It generates software tests, predicts pharmaceutical batch failures, detects financial model drift, recommends agronomic interventions, retrieves consulting methodologies, and optimises research experiment designs — all from one intelligence engine with full lifecycle context for every solution it manages.

Success Metrics: AI recommendation acceptance rate > 60% within 12 months. Defect/failure escape rate reduced by 35%+ across all solution types. AI becomes the organisation's solution intelligence engine, not just a coding assistant.

OPP-009 — Extend Software-Grade Infrastructure to the Entire Economy

The software industry has spent 30 years building world-class delivery infrastructure. Only software-native organisations benefit. Qala takes the same principles — version control, reproducible builds, governed release, continuous testing, immutable audit — and makes them universally available. A farmer, a pharmacist, a fund manager, and a freelance consultant all get the same infrastructure quality as a FAANG engineering team.

Success Metrics: Qala's SAM is 10× larger than the DevOps tools market alone. Non-software solution lifecycle management is a $200B+ untapped opportunity.

5 Problem–Solution Fit Matrix

Every identified problem is mapped to the specific Qala capability that resolves it, the solution types it affects, and the primary success metric.

Problem ID	Problem Statement	Qala Capability	Solution Types	Success Metric
P-SW-001	Non-reproducible builds & environment inconsistency	Hermetic SDE + env-as-code + digest-pinned dependencies	All software types	Zero environment-caused CI failures
P-SW-002	Security bolted on, not built in	SAST/SCA/DAST in every pipeline + SEM + signed attestations	All software types	95%+ vulnerabilities caught in development phase
P-SW-003	Ungoverned release and deployment	8-gate release management + Blue/Green/Canary deployment	All software types	Change failure rate < 5%; MTTR < 1 hour
P-SW-004	Tool/library lifecycle ungoverned	Tool Registry + governed toolchain + SBOM generation	Tools, Libraries, Packages, SDKs, Toolchains	Zero unscanned tools in production; all consumers notified on breaking change
P-GOOD-001	Formulation/specification version chaos	Solution Model: Good/CPG + CCR workflow + version control	Goods, CPG, Products, Product Lines	Single authoritative version; 100% changes in CCR workflow
P-GOOD-002	Recall readiness absent	Artefact Registry + batch traceability + distribution events	Goods, CPG, Food, Pharma, Medical Devices	Affected batch identification in < 5 minutes
P-GOOD-003	Product line governance across variants impossible	Product Line solution type + component-level CCR propagation	Product Lines, CPG portfolios	Impact assessment for component change in < 1 hour
P-FIN-001	Financial product lifecycle ungoverned	Versioned product specs + CCR workflow + compliance reporting	Financial Instruments, Investment Products, Insurance	Zero undocumented financial product changes; MiFID II audit trail complete
P-FIN-002	Capital/investment tracking across vehicles	Investment solution type + mandate versioning + model registry	Investment Solutions, Capital Solutions, Funds	100% of mandates versioned; model drift detected within 1 day
P-FIN-003	Tax solution changes manual and risky	Tax Solution type + effective-date release + rule versioning	Tax Solutions, Regulatory Products	Zero missed effective dates; all rule changes in CCR workflow

P-AGRI-001	Farm-to-consumer traceability broken	Event architecture + provenance chain + distribution events	Agricultural Solutions, Food Supply Chain	Full trace from consumer to field in < 10 minutes
P-AGRI-002	Agricultural solutions not lifecycle-managed	Agricultural SDE + seasonal release cadence + agri AI model	Agricultural Platforms, AgTech, Crop Systems	100% agri-solutions on governed lifecycle; firmware OTA via pipeline
P-SVC-001	Service knowledge trapped in individuals	Knowledge SDE + versioned methodology + AI semantic retrieval	Services, Consulting, Coaching, Professional Services	Any methodology retrieved in < 2 minutes; zero knowledge loss on personnel change
P-SVC-002	Managed service specs not governed	Service solution type + version control + SLA gate	Managed Services, Subscription Services	100% service specification changes in CCR workflow
P-BIZ-001	Business solutions have no post-launch lifecycle	Business Solution type + process version control + outcome benchmarking	Business Solutions, Process Solutions, Operational Solutions	As-designed vs. as-operated drift detected continuously; outcome metrics tracked
P-RES-001	Research not reproducible	Hermetic research SDE + attestation + dataset versioning	Research Assets, Datasets, Experimental Frameworks	100% of computational experiments reproducible from artefact record
P-RES-002	Organisational resources have no lifecycle	Resource/Asset solution type + governed deprecation + consumer notification	Resources, Assets, Templates, Frameworks	All consumers notified on resource change; zero deprecated templates in active use

6 Success Metrics — By Domain

Success is measured not by features shipped but by whether the problems are genuinely solved. Each domain has specific, quantified outcome metrics.

6.1 Software and Technology

Metric	Industry Baseline	Qala Target
SDE provisioning time	2–14 days (manual setup)	< 10 minutes — automated from template
Build reproducibility rate	< 30% of organisations	100% — hermetic by default
Vulnerability detection phase (dev vs. prod)	40% in dev, 60% in prod	> 95% in development phase
CI pipeline security gate coverage	< 40% of pipelines	100% — mandatory in platform default
Deployment frequency (DORA)	Monthly (industry median)	Multiple per day (elite performer level)
Change failure rate (DORA)	15–45%	< 5%
MTTR from production incident (DORA)	1 day – 1 week	< 1 hour (5-minute rollback always available)
Onboarding time to productive contribution	1–4 weeks	< 1 business day

6.2 Physical Goods, CPG and Manufacturing

Metric	Industry Baseline	Qala Target
Time to identify affected batches in recall	2–14 days manual	< 5 minutes via registry query
Formulation change audit completeness	30–60%	100% — all changes in CCR workflow
Product line impact assessment time	1–4 weeks manual	< 1 hour via governed component relationships
Regulatory submission rejection rate	15–25%	< 3% — automated compliance pre-check before submission
Specification version discrepancies in production	Common	Zero — single authoritative version enforced

6.3 Financial and Investment Solutions

Metric	Industry Baseline	Qala Target
Financial product change audit trail completeness	< 40%	100% — immutable CCR records

Compliance preparation time (SOC2/MiFID II)	4–12 weeks	< 4 hours — automated evidence generation
Model drift detection time	Days to never	< 24 hours via AI monitoring
Regulatory examination finding rate (governance)	Industry average	80%+ reduction in governance-related findings
Effective-date release success rate for tax products	75% on first attempt	> 99% — governed effective-date pipeline

6.4 Agricultural and Agribusiness

Metric	Industry Baseline	Qala Target
Farm-to-consumer traceability time	2–7 days manual	< 10 minutes via provenance chain query
Governed agri-solution lifecycle adoption	< 5% of AgTech solutions	100% of solutions onboarded to platform have governed lifecycle
IoT firmware OTA update governance	Largely ungoverned	100% OTA updates via governed release pipeline with attestation
Yield improvement from AI intelligence	Baseline	15–25% yield improvement from unified data and AI recommendations

6.5 Professional Services and Consulting

Metric	Industry Baseline	Qala Target
Knowledge retrieval time for methodology	30 min – 4 hours	< 2 minutes via AI semantic search
Knowledge loss on senior departure	High — 30–50% dark knowledge	Near-zero — all intellectual capital versioned and AI-indexed
Onboarding time for new consultant	1–3 months	< 1 week — knowledge base immediately accessible
Engagement methodology reuse rate	20–40%	> 75% with AI-recommended methodology matching
Service specification change traceability	< 20% of changes documented	100% in CCR workflow

6.6 Platform Adoption Targets

Metric	Target: 12 Months	Target: 36 Months
Solution types actively managed on platform	8 of 40+ defined types	All 40+ types with vertical-specific workflows
Active solution records	100,000	10,000,000+
Industries with active deployments	8+	25+

Non-software solutions as % of total	10%	40%
AI recommendation acceptance rate	> 55%	> 75%
Automated compliance evidence coverage	> 70% of supported frameworks	> 99%
Platform NPS	> 45	> 65

7 Constraints, Assumptions and Risks

7.1 Design Constraints

Constraint	Description
Universal Solution Model Coherence	The solution type taxonomy and universal attribute model must coherently represent all identified solution types without forcing unnatural conceptual mappings. Any solution type that cannot be coherently represented must be explicitly deferred.
Industry-Neutral Platform Core	All domain-specific behaviour must be delivered through the Domain Extension framework — configurable schemas, vertical workflow templates, compliance adapters. The platform core must have zero hardcoded industry logic.
Open Standards Only	Every interface, data format, and protocol must use open standards: OpenAPI, AsyncAPI, SPDX, CycloneDX, SLSA, OAIS, OpenTelemetry, OAuth2/OIDC, GS1 EPCIS, FIX protocol adapters, and equivalents. No proprietary formats.
Physical World Boundary	Qala manages the information layer for physical solutions — specifications, versions, quality records, distribution records. It does not manage the physical production process itself. Integration with ERP, MES, LIMS, and physical systems is via open APIs.
Non-Software UX Parity	A CPG Product Developer, a Farm Operations Manager, and a Tax Product Manager must find Qala as intuitive and purpose-built for their domain as a software developer does. Non-software users must never encounter software-native concepts unless directly relevant to their work.
Compliance Modularity	Compliance frameworks are modules. Adding support for a new regulatory regime requires configuration, not code changes. The compliance engine processes evidence mappings dynamically from compliance pack definitions.
Performance at Universal Scale	The platform must sustain 10M+ active solution records across all types, with heterogeneous schema sizes and event frequencies, without degradation of the API SLA (< 500ms p95).

7.2 Strategic Risks

Risk	Likelihood	Impact	Mitigation
Platform scope perceived as too broad	High	High	Lead with software and 2–3 adjacent verticals at launch. Expand via a validated Vertical Release Programme. "Universal by design, specific in delivery."
Non-software UX alienates domain experts	High	High	Dedicated UX research per non-software persona. Role-specific portals. Domain vocabulary overlays. Avoid surfacing CI/CD concepts to a CPG Product Developer.
Solution model too abstract for practitioners	Medium	High	Co-design solution type schemas with domain experts before implementation. Beta programmes per vertical with design partners.
AI insufficient quality for regulated domains	High	Medium	AI features launched per domain as training data matures. No AI-automated actions in regulated domains without explicit human approval and audit trail.

Regulatory compliance modules become stale	Medium	High	Partner with regulatory intelligence providers. Automated regulatory change monitoring. Rapid compliance pack update process (< 30 days from regulation change to pack update).
Physical world integration complexity underestimated	Medium	High	Define the platform boundary clearly: Qala manages information. Open API integration spec for ERP/MES/LIMS. Integration packs for SAP, Oracle, Siemens MES.
Supply chain attack on Qala itself	Low	Critical	Qala is built with Qala. Platform infrastructure managed as a solution in Qala with hermetic builds, SLSA attestation, and all security gates applied to the platform itself.

7.3 Prioritised Problem Backlog

Rank	Problem ID	Problem	Severity	Breadth	P
1	P-SW-003	Ungoverned release and deployment	Critical	Universal	P1
2	P-SW-001	Non-reproducible builds and environment inconsistency	High	Universal	P1
3	P-SW-002	Security bolted on, not built in	High	Universal	P1
4	P-GOOD-002	Recall readiness absent	Critical	CPG/Pharma/Food	P1
5	P-GOOD-001	Formulation and specification version chaos	High	CPG/Mfg/Pharma	P1
6	P-FIN-001	Financial product lifecycle ungoverned	High	Financial	P1
7	P-SW-004	Tool/library lifecycle ungoverned	High	Universal	P1
8	P-AGRI-001	Farm-to-consumer traceability broken	High	Agri/Food	P1
9	P-BIZ-001	Business solutions have no post-launch lifecycle	High	Enterprise	P2
10	P-SVC-001	Service knowledge trapped in individuals	High	Services	P2
11	P-RES-001	Research not reproducible	High	Research/ Pharma	P2
12	P-FIN-003	Tax solution changes manual and risky	High	Tax/Legal	P2
13	P-AGRI-002	Agricultural solutions not lifecycle-managed	Medium	AgTech	P2
14	P-GOOD-003	Product line governance across variants impossible	Medium	CPG/Mfg	P2

15	P-FIN-002	Capital/investment tracking across vehicles	Medium	Asset Mgmt	P2
16	P-SVC-002	Managed service specs not governed	Medium	MSP/SaaS	P3
17	P-RES-002	Organisational resources have no lifecycle	Medium	Universal	P3

End of Document — Qala Universal Solution Management OS | Problem Requirements Document v2.0